**HW7**

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## Part a. Estimation Results

```
US alpha constraint: 0.6261774125925925
US ln(Z_bar) Estimate: 2.785950673828649
P-value: 1.826347395110573e-52
UK alpha constraint: 0.6109429948148148
UK ln(Z_bar) Estimate: 2.6542584132482943
P-value: 6.560628363331106e-60
```

```
37]: print(results.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             dep_var_uk   R-squared:                       0.000
Model:                            OLS   Adj. R-squared:                  0.000
Method:                 Least Squares   F-statistic:                       nan
Date:                Sat, 25 Apr 2026   Prob (F-statistic):                nan
Time:                        15:05:12   Log-Likelihood:                 7.0832
No. Observations:                  54   AIC:                            -12.17
Df Residuals:                      53   BIC:                            -10.18
Df Model:                           0
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      2.6543      0.029     91.051      0.000       2.596       2.713
==============================================================================
Omnibus:                       23.604   Durbin-Watson:                   0.011
Prob(Omnibus):                  0.000   Jarque-Bera (JB):                4.044
Skew:                           0.005   Prob(JB):                        0.132
Kurtosis:                       1.659   Cond. No.                         1.00
==============================================================================
```

For each country, I estimated the production function scaling constant $\overline{Z}^i$ using a constrained regression, where the labor share $\alpha^i$ was fixed at the sample mean of the variable labsh_i. The residuals from these regressions were saved as log_us_zpop and log_uk_zpop respectively.

**US:**The estimate for $\ln(\overline{Z^{US}})$ is **2.785951**.

The estimate for $\overline{Z^{US}}$ is $16.21$.

This estimate is **statistically significant at the 1 percent level** (p-value < 0.01).

**UK:**The estimate for $\ln(\overline{Z^{UK}})$ is **2.654258**

The estimate for $\overline{Z^{UK}}$ is 14.21.

This estimate is **statistically significant at the 1 percent level** (p-value < 0.01).

## Part b. VAR Analysis and Spillovers

### 1. Optimal Lag Selection

To determine the dynamic relationship between the productivity shocks of the US and the UK, I first conducted a lag-order selection test using information criteria (AIC, BIC, HQIC, and FPE). Based on the results, **N=1** was chosen as the optimal lag length, as it was the shortest lag suggested by the majority of the criteria, ensuring model simplicity while capturing the essential dynamics.

VAR Order Selection (* highlights the minimums)

| | AIC | BIC | FPE | HQIC |
|---|---|---|---|---|
| 0 | -10.09 | -10.01 | 4.139e-05 | -10.06 |
| 1 | -16.37* | -16.13* | 7.754e-08* | -16.28* |
| 2 | -16.32 | -15.91 | 8.195e-08 | -16.17 |
| 3 | -16.19 | -15.62 | 9.354e-08 | -15.98 |
| 4 | -16.07 | -15.34 | 1.062e-07 | -15.80 |
| 5 | -16.01 | -15.12 | 1.137e-07 | -15.68 |
| 6 | -15.99 | -14.94 | 1.179e-07 | -15.60 |
| 7 | -15.86 | -14.64 | 1.376e-07 | -15.40 |
| 8 | -15.82 | -14.44 | 1.468e-07 | -15.31 |
| 9 | -15.94 | -14.40 | 1.350e-07 | -15.37 |
| 10 | -15.89 | -14.19 | 1.487e-07 | -15.26 |

Summary of Regression Results

| | | | |
|---|---|---|---|
| Model: | VAR | | |
| Method: | OLS | | |
| Date: | Sat, 25, Apr, 2026 | | |
| Time: | 15:28:10 | | |
| No. of Equations: | 2.00000 | BIC: | -16.1745 |
| Nobs: | 53.0000 | HQIC: | -16.3118 |
| Log likelihood: | 290.127 | FPE: | 7.56394e-08 |
| AIC: | -16.3975 | Det(Omega_mle): | 6.77523e-08 |

Results for equation log_us_zpop

| | coefficient | std. error | t-stat | prob |
|---|---|---|---|---|
| const | 0.018939 | 0.002124 | 8.915 | 0.000 |
| L1.log_us_zpop | 1.001897 | 0.037368 | 26.812 | 0.000 |
| L1.log_uk_zpop | -0.041205 | 0.054444 | -0.757 | 0.449 |

Results for equation log_uk_zpop

| | coefficient | std. error | t-stat | prob |
|---|---|---|---|---|
| const | 0.011015 | 0.002506 | 4.395 | 0.000 |
| L1.log_us_zpop | 0.156165 | 0.044079 | 3.543 | 0.000 |
| L1.log_uk_zpop | 0.768823 | 0.064222 | 11.971 | 0.000 |

Correlation matrix of residuals

| | log_us_zpop | log_uk_zpop |
|---|---|---|
| log_us_zpop | 1.000000 | 0.384622 |
| log_uk_zpop | 0.384622 | 1.000000 |

**2. VAR Estimation Results**

The estimated VAR(1) model results are presented in the table below:

| Statistic | Value | Significant at |
|---|---|---|
| $\chi^{US}$ point estimate | **0.0189** | **1%** |
| $\chi^{UK}$ point estimate | **0.0110** | **1%** |
| $\rho^{1,US}$ point estimate | **1.0019** | **1%** |
| $\rho^{1,UK}$ point estimate | **0.7688** | **1%** |
| $\phi^{1,US}$ point estimate | **-0.0412** | **NS** |
| $\phi^{1,UK}$ point estimate | **0.1562** | **1%** |
| variance of eUS | **0.000057** | N/A |
| variance of eUK | **0.000156** | N/A |
| correlation of eUS with eUK | **0.3846** | N/A |

**3. Interpretation of Granger Causality and Ordering**

The Granger causality test, derived from the VAR coefficients, reveals a **unidirectional relationship** between the two economies:

**US to UK:** The coefficient $\phi^{1,UK}$ (0.1562) is highly significant (p = 0.000), indicating that productivity shocks in the US strongly predict future productivity in the UK.

**UK to US:** The coefficient $\phi^{1,US}$ (-0.0412) is statistically insignificant (p = 0.449), suggesting that UK shocks have no predictive power over US productivity.

Given this evidence, I chose the variable ordering as **[US, UK]**. This recursive structure implies that the US acts as the exogenous source, while the UK responds to these shocks with a lag.

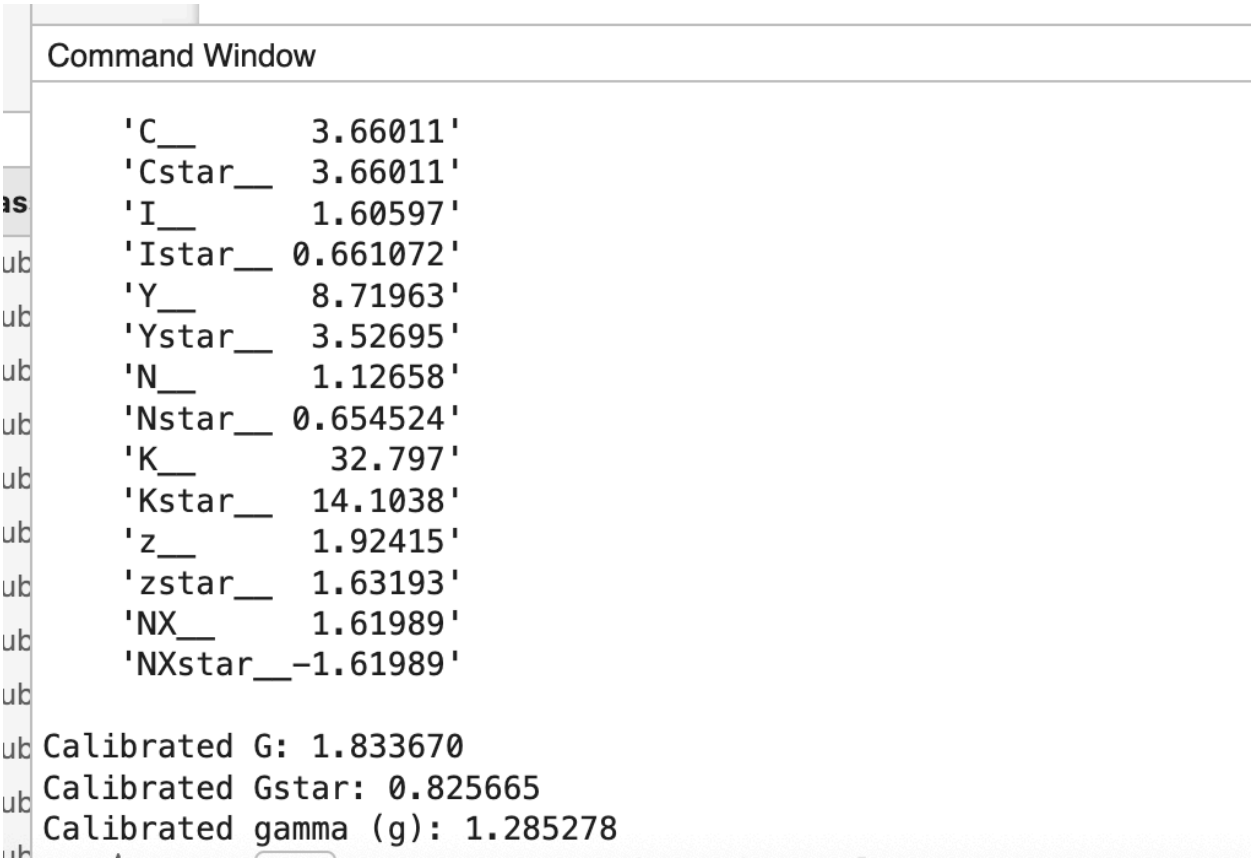
**4. Spillovers**

The results provide strong evidence of **positive spillovers** from the US to the UK. A 1% increase in US productivity is estimated to lead to a 0.156% increase in UK productivity in the following period. This reflects the role of the US as a global frontier, where innovations originated in the US eventually benefit the UK economy. Conversely, there is no evidence of a reverse spillover from the UK to the US, as the relevant coefficient is small and statistically indistinguishable from zero.

## Part c.

## See the part c.PDF document

## Part d.




| Parameter | Value |
|---|---|
| G | 1.833670 |
| G* | 0.825665 |
| g | 1.285278 |

## Part e.

**Missing parameter values：** g = 1.285278; G = 1.83367; Gstar = 0.82566;

**Missing equation:**

```
log(z) = chi + rho1*log(z(-1)) + rho1star*log(zstar(-1)) + ez;
log(zstar) = chistar + phi1*log(z(-1)) + phi1star*log(zstar(-1)) + ezstar;
NX = Y - C - I - G;
NXstar = Ystar - Cstar - Istar - Gstar;
```

**Initval:**

```
C = 3.66011;
Cstar = 3.66011;
I = 1.60597;
Istar = 0.66107;
Y = 8.71963;
Ystar = 3.52695;
N = 1.12658;
Nstar = 0.65452;
K = 32.7970;
Kstar = 14.1039;
z = 1.92415;
zstar = 1.63193;
NX = 1.61989;
NXstar = -1.61989;
```

